

Integrating Regularized Covariance Matrix Estimators

Clifford Lam* and Phoenix Feng†

Department of Statistics, London School of Economics and Political Science

Abstract

Covariance regularization is important when the dimension p of a covariance matrix is close to or even larger than the sample size n . One branch of regularization assumes specific structures, like sparse, banded, or a having a factor structure for the true covariance matrix Σ_0 . Another branch regularize on the eigenvalues directly without assuming these structures. Under the practical scenario where one is not 100% certain of which regularization method to use, we introduce an integration covariance matrix estimator which is a linear combination of a rotation-equivariant and a regularized covariance matrix estimator that assumed a specific structure for Σ_0 . We estimate the weights in the linear combination and show that they asymptotically go to the true underlying weights. To generalize, we can put two regularized estimators into the linear combination, each assumes a specific structure for Σ_0 . Our estimated weights can then be shown to go to the true weights too, and if one regularized estimator is converging to Σ_0 in the spectral norm, the corresponding weight then tends to 1 and others tend to 0 asymptotically. We demonstrate the performance of our estimator when compared to other state-of-the-art estimators through extensive simulation studies and a real data analysis.

MSC classification : 62G20, 15B52

Key words and phrases. High dimensional data analysis, integration of regularized estimators, data splitting, nonlinear shrinkage, Frobenius loss.

*Clifford Lam is Associate Professor, Department of Statistics, London School of Economics. Email: C.Lam2@lse.ac.uk

†Phoenix Feng is PhD student, Department of Statistics, London School of Economics. Email: H.Feng2@lse.ac.uk

1 Introduction

Estimation of a covariance matrix or its inverse, called the precision matrix, is an important and sometimes inevitable task in data analysis. Examples range from risk estimation and portfolio allocation in finance to classification or multiple hypotheses testing in general statistical analysis. Although it is easier than ever nowadays to obtain relatively large data set for study, such richness of data also means that more often than not the data we obtain are high dimensional in nature, in the sense that the number of variables under study is comparable to or even larger than the sample size. This creates problems for traditional estimator such as the sample covariance matrix. Well-documented in Bai and Yin (1993) and subsequent random matrix theory researches (see Bai and Silverstein (2010), for example), a particularly serious problem is that the eigenvalues of the sample covariance matrix are more extreme than their population counterpart. Moreover, when the dimension p is larger than the sample size n , the sample covariance matrix is not invertible.

In view of these problems, two major branches of regularization methods are developed for covariance matrix estimation. One branch assumes that the population covariance matrix Σ_0 or the corresponding precision matrix Σ_0^{-1} has special structures. Sparseness of Σ_0 (i.e., many zeros or near-zeros in Σ_0) is one of the most commonly exploit structure in applications (Bickel and Levina, 2008a, Cai and Yuan, 2012, Cai and Zhou, 2012, Lam and Fan, 2009, Rothman et al., 2009). Sparseness of Σ_0^{-1} is closely connected to graphical modelling (Friedman et al., 2008, Meinshausen and Bühlmann, 2006). Banded structure of Σ_0 and Σ_0^{-1} (i.e., only limited number of off-diagonals and the main diagonal are non-zero) are treated in Bickel and Levina (2008b). If the data follows a factor model, then Σ_0 can have a factor structure (Fan et al., 2008, 2011). Fan et al. (2013) combines factor structure estimation as well as sparseness exploitation in the residual covariance matrix.

Another branch concerns with regularizing, or “shrinking”, the eigenvalues of the sample covariance matrix under the high dimensional setting $p/n \rightarrow c > 0$. Ledoit and Wolf (2004) proposed the linear shrinkage estimator, which is a weighted average of the identity and the sample covariance matrix, shrinking the eigenvalues towards a grand mean. This is generalized to shrinkage towards a specified matrix “target” other than the identity in Schäfer and Strimmer (2005). Won et al. (2013) regularized on the condition number, winsorizing the extreme eigenvalues of the sample covariance matrix at certain constants. Ledoit and Wolf (2012, 2013) proposed a rotation-equivariant estimator with nonlinear shrinkage of eigenvalues, while Lam (2016), using a data-splitting idea from Abadir et al. (2014), proved that such nonlinear shrinkage can be achieved through data-splitting with theoretical justification of the split location. Recently, Huang and Fryzlewicz (2015) proposed the NOVELIST, a weighted average of the sample covariance matrix and a thresholded estimator, and obtained good practical results. This can also be considered as a form of nonlinear shrinkage estimator, although there are no theoretical justifications in the exact form of shrinkage.

While both branch of researches provide good estimators under different scenarios, there are no methods that can “take advantage” of what each branch of estimators can offer. For instance, if the sparse assumption on Σ_0 is correct, then a thresholded estimator $\mathbf{T}$ is good. Yet, $\mathbf{T}$ can be far from Σ_0 if the sparse assumption turns out to be only roughly true or not true at all. Even when the sparse assumption is correct, finite sample performance of $\mathbf{T}$ can still be not as good as a shrinkage estimator. With this in mind, a desirable estimator is one that can be automatically similar to a structured estimator $\mathbf{T}$ when it is indeed close to

Σ_0 in a certain sense, and be similar to a shrinkage estimator automatically when $\mathbf{T}$ is not performing well. Moreover, when Σ_0 is approximately banded and approximately sparse for instance, we hope that a regularized covariance matrix estimator can take advantages from both a banded and a thresholded estimator at the same time, and be able to “switch” to one particular estimator if that is close enough to Σ_0 in a certain sense.

To bridge the gap described above, we propose an estimator that combines two or even three regularized covariance estimators through a weighted average. This is not dissimilar with the NOVELIST in Huang and Fryzlewicz (2015), except that we are combining a rotation-equivariant estimator, instead of just the sample covariance matrix, with other regularized estimators. The freedom of the diagonals in the rotation-equivariant estimator, together with the weight on each regularized estimator, allows for a flexible final estimator. One major contribution in this paper is that, by minimizing the Frobenius loss, we derive explicit formulae for the weights which can be estimated consistently, or even almost surely, through a data splitting scheme which is similar to the one in Lam (2016). These weights indeed allow our estimator to have the desirable property that it approaches a particular regularized estimator $\mathbf{T}$ if it is indeed close to Σ_0 in a certain sense, and approaches the rotation-equivariant estimator if other regularized estimators are not performing well. See Section 2 and Theorem 2 and 5 for more details. Extensive simulations also reveal that our estimator can indeed gather the advantages from a regularized estimator and perform well, even when the regularized estimator itself does not. As another contribution, we show that our estimator is asymptotically efficient when compared to an ideal estimator constructed with the knowledge of Σ_0 . Such efficiency proof, similar to Lam (2016), also provides possible theoretical split locations, and give insights into choosing a practical one for the data. Finally, extension to even more regularized estimators is not difficult using the mathematics behind the proof of Theorem 1 and 4, although we do not pursue in this direction in the paper.

The rest of the paper is organized as follows. Section 2 defines the concept of our estimator, and shows the form of an ideal estimator through Frobenius loss minimization. A bona fide estimator is introduced in Section 2.2 with data splitting, and its theoretical properties presented in Section 2.3. Section 3 extends everything to two regularized covariance matrices. Section 4.1 introduces an averaged estimator which has better performance and is more stable, while detailing the practical procedures in the choice of split location for the data and other practical concerns. Section 5 presents the extensive simulation results with a real data analysis as well.

2 Proposed Estimator with a Single Regularized Matrix

We consider a covariance matrix estimator of the form

$$\Sigma(\delta, \mathbf{D}) = (1 - \delta)\mathbf{PDP}^\top + \delta\mathbf{T}, \quad (2.1)$$

where $\mathbf{T}$ is a regularized covariance matrix using a chosen regularization method, and $\mathbf{P}$ is an orthogonal matrix. The above estimator is weighted between the regularized estimator $\mathbf{T}$ and the matrix $\mathbf{PDP}^\top$. Lam (2016) used a rotation-equivariant estimator $\mathbf{PDP}^\top$ as the basis of a covariance matrix estimator, where $\mathbf{P}$ contains all the eigenvectors of a sample covariance matrix constructed from the available data. Similar to Abadir et al. (2014), a sample splitting scheme is used in Lam (2016) to find the diagonal matrix $\mathbf{D}$.

Our estimator closely resembles the NOVEL Integration of the Sample and Thresholded covariance estimators (NOVELIST) proposed in Huang and Fryzlewicz (2015), in the sense that our estimator also aims to integrate with a regularized estimator, for instance, a thresholded covariance matrix estimator $\mathbf{T}$, using a linear weighting scheme. As such, if $\mathbf{T}$ is a fixed “target” matrix which does not depend on data, it also resembles the linear shrinkage estimator proposed in Ledoit and Wolf (2004), where a linear combination of the sample covariance matrix and a fixed target matrix is considered.

The major difference, however, is that the sample covariance matrix in both papers is replaced by a rotation-equivariant estimator $\mathbf{PDP}^\top$, where $\mathbf{D}$ is diagonal matrix to be determined. Hence, we are integrating two regularized covariance matrices, instead of just regularizing the sample covariance matrix through linear weighting with another regularized estimator. The major motivation in doing so is that while the rotation-equivariant estimator can help achieve nonlinear shrinkage of eigenvalues as in Lam (2016) without assuming a specific structure of the true covariance matrix Σ_0 , it may lose accuracy against a regularized estimator which assumes a specific structure for Σ_0 , if such an assumption does ultimately hold. In this sense, we hope to achieve a better estimator through $\Sigma(\delta, \mathbf{D})$ when we are not certain if a particular assumption on Σ_0 holds. Ideally, the weight δ should be close to 1 if $\mathbf{T}$ is regularized correctly assuming a specific structure on Σ_0 , and close to 0 if $\mathbf{T}$ is too far off the true one because of a wrong assumption on Σ_0 . We show that this is indeed the case asymptotically almost surely using our proposed estimator (2.5) in Section 2.2 below.

2.1 Frobenius loss minimization

We propose to estimate δ and $\mathbf{D}$ through minimizing the Frobenius loss. Ledoit and Wolf (2004) also used Frobenius loss minimization for their linear shrinkage estimator. Hence, we consider

$$\min_{\delta, \mathbf{D}} \|(1 - \delta)\mathbf{PDP}^\top + \delta\mathbf{T} - \Sigma_0\|_F^2, \quad (2.2)$$

where $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}\mathbf{A}^\top)$, with $\text{tr}(\mathbf{A})$ being the trace of a matrix $\mathbf{A}$. We present the solution to the minimization problem above in the following theorem.

Theorem 1. *Suppose $\delta \neq 1$ and $\mathbf{T}$ is not of the form $\mathbf{PDP}^\top$ for some diagonal matrix $\mathbf{D}$. Define $\hat{\Sigma}_{\mathbf{T}} = \mathbf{P}\text{diag}(\mathbf{P}^\top\mathbf{T}\mathbf{P})\mathbf{P}^\top$, where $\text{diag}(\mathbf{A})$ represents a diagonal matrix with diagonal entries as in the matrix $\mathbf{A}$. Then the solution to the minimization problem (2.2) is*

$$\mathbf{D} = \frac{1}{1 - \delta}\text{diag}(\mathbf{P}^\top\Sigma_0\mathbf{P}) - \frac{\delta}{1 - \delta}\text{diag}(\mathbf{P}^\top\mathbf{T}\mathbf{P}), \text{ with } \delta = \frac{\text{tr}[(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})\Sigma_0]}{\text{tr}(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})^2}.$$

The requirement that $\mathbf{T}$ cannot be of the form $\mathbf{PDP}^\top$ is a regularity condition. If this condition is not satisfied, then $\text{tr}(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})^2 = 0$, and the original problem actually reduces to the one considered in Lam (2016) or Ledoit and Wolf (2012). The same thing happens if $\Sigma_0 = \sigma^2\mathbf{I}_p$, since then $\text{tr}[(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})\Sigma_0] = 0$, and so $\delta = 0$. Substituting the forms of $\mathbf{D}$ and δ into $\Sigma(\delta, \mathbf{D})$, the corresponding covariance matrix estimator depends on $\mathbf{P}$, $\mathbf{T}$ and Σ_0 , and is given by

$$\Sigma(\mathbf{P}, \mathbf{T}, \Sigma_0) = \mathbf{P}\text{diag}(\mathbf{P}^\top\Sigma_0\mathbf{P})\mathbf{P}^\top + \frac{\text{tr}[(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})\Sigma_0]}{\text{tr}(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})^2}(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}}). \quad (2.3)$$

The first part of the estimator coincides with the “ideal” nonlinear shrinkage estimator in Ledoit and Wolf (2012) and Lam (2016) for efficiency comparisons purpose. The second part is a weighted version of $\mathbf{T} - \hat{\Sigma}_{\mathbf{T}}$, which is itself congruent to a hollow matrix, in the sense that $\mathbf{P}(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})\mathbf{P}^T$ has zero diagonal. The weight $\text{tr}[(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})\Sigma_0]/\text{tr}(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})^2$ can actually go negative or above 1. Looking at its formula, it is a generalized angle between Σ_0 and $\mathbf{T} - \hat{\Sigma}_{\mathbf{T}}$, calculated using the inner product $\text{tr}(\mathbf{AB})$ for symmetric real square matrices $\mathbf{A}$ and $\mathbf{B}$. When the inner product $\text{tr}[(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})\Sigma_0] = 0$, it means that Σ_0 and $\mathbf{T} - \hat{\Sigma}_{\mathbf{T}}$ are “orthogonal” to each other in a generalized sense, and hence $\mathbf{T} - \hat{\Sigma}_{\mathbf{T}}$ is not important and is therefore weighted 0, leaving only the ideal nonlinear shrinkage estimator $\mathbf{P}\text{diag}(\mathbf{P}^T\Sigma_0\mathbf{P})\mathbf{P}^T$. Obviously we cannot use this estimator in practice since it depends on Σ_0 itself.

Remark 1. Theorem 1 does not allow $\delta = 1$. However, as $\delta \rightarrow 1$, using $\text{tr}[(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})\hat{\Sigma}_{\mathbf{T}}] = 0$, we can easily deduce from the form of δ in the result of Theorem 1 that the estimator $\Sigma(\mathbf{P}, \mathbf{T}, \Sigma_0)$ approaches the form $\mathbf{T} + \mathbf{P}\text{diag}(\mathbf{P}^T(\Sigma_0 - \mathbf{T})\mathbf{P})\mathbf{P}^T$. This makes sense, since even if $\mathbf{T}$ is a good estimator in itself so that we assume $\delta \rightarrow 1$ in (2.1), in terms of minimizing the Frobenius loss, $\mathbf{T} + \mathbf{P}\text{diag}(\mathbf{P}^T(\Sigma_0 - \mathbf{T})\mathbf{P})\mathbf{P}^T$ is always better than $\mathbf{T}$ alone.

2.2 Proposed estimator with data splitting

Since Σ_0 is unknown, we need to propose a practical estimator which can be constructed from the data $\mathbf{Y} = (\mathbf{y}_1, \dots, \mathbf{y}_n)$, where each observed vector $\mathbf{y}_i$ is of dimension p . Hence Σ_0 is a $p \times p$ matrix. Hereafter we assume that the $\mathbf{y}_i$ ’s are independent of each other, each with mean $\mathbf{0}$ and covariance Σ_0 .

We use the sample splitting idea used in Abadir et al. (2014) and Lam (2016), and split the data into two independent portions $\mathbf{Y} = (\mathbf{Y}_1, \mathbf{Y}_2)$, where $\mathbf{Y}_1$ is of size $p \times m$ and $\mathbf{Y}_2$ of size $p \times n - m$. Construct two sample covariance matrices, and perform an eigen-decomposition on the first one:

$$\tilde{\Sigma}_1 = m^{-1}\mathbf{Y}_1\mathbf{Y}_1^T = \mathbf{P}_1\mathbf{D}_1\mathbf{P}_1^T, \quad \tilde{\Sigma}_2 = (n - m)^{-1}\mathbf{Y}_2\mathbf{Y}_2^T.$$

We then consider the Frobenius loss minimization problem

$$\min_{\delta, \mathbf{D}} \|(1 - \delta)\mathbf{P}_1\mathbf{D}\mathbf{P}_1^T + \delta\mathbf{T} - \Sigma_0\|_F^2, \quad (2.4)$$

where $\mathbf{T}$ here is constructed from $\mathbf{Y}_1$ only instead of the whole data matrix $\mathbf{Y}$. This results in the estimator $\Sigma(\mathbf{P}_1, \mathbf{T}, \Sigma_0)$ similar to that in (2.3). We then propose our estimator to be $\Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2)$, that is,

$$\Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2) = \mathbf{P}_1\text{diag}(\mathbf{P}_1^T\tilde{\Sigma}_2\mathbf{P}_1)\mathbf{P}_1^T + \frac{\text{tr}[(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})\tilde{\Sigma}_2]}{\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}}). \quad (2.5)$$

Note here $\tilde{\Sigma}_{\mathbf{T}} = \mathbf{P}_1\text{diag}(\mathbf{P}_1^T\mathbf{T}\mathbf{P}_1)\mathbf{P}_1^T$. The first part of this estimator is exactly the NERCOME introduced in Lam (2016) for nonlinear shrinkage of the eigenvalues of the sample covariance matrix. The salient feature that gives this estimator its nice properties (to be introduced in later theorems) is that both $\mathbf{P}_1$ and $\mathbf{T}$ are independent of $\tilde{\Sigma}_2$ by our construction. For the theoretical value of the split m and a practical one, see Section 2.3 and Section 4 respectively.

2.3 Theoretical results with single regularized estimator

To present the asymptotically properties of our estimator $\Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2)$, we list our assumptions below.

- (A1) The observed vectors are independent of each other, and each $\mathbf{y}_i$ can be written as $\mathbf{y}_i = \Sigma_0^{1/2} \mathbf{z}_i$ for $i = 1, \dots, n$, where each $\mathbf{z}_i$ is a $p \times 1$ vector of independent and identically distributed random variables z_{ij} . Each z_{ij} has mean 0 and variance 1, with $E|z_{ij}|^k \leq B < \infty$ for some constant B and $2 < k \leq 20$.
- (A2) The population covariance matrix is non-random and of size $p \times p$. Furthermore, $\|\Sigma_0\| = O(1)$.
- (A3) Let $\tau_{n,1} \geq \dots \geq \tau_{n,p}$ be the p eigenvalues of Σ_0 , with corresponding eigenvectors $\mathbf{v}_{n,1}, \dots, \mathbf{v}_{n,p}$. Define $H_n(\tau) = p^{-1} \sum_{i=1}^p \mathbf{1}_{\{\tau_{n,i} \leq \tau\}}$ the empirical distribution function (e.d.f.) of the population eigenvalues, where $\mathbf{1}_A$ is the indicator function of the set A . We assume $H_n(\tau)$ converges to some non-random limit H at every point of continuity of H .
- (A4) The support of H defined above is the union of a finite number of compact intervals bounded away from zero and infinity. Also, there exists a compact interval in $(0, +\infty)$ that contains the support of H_n for each n .

For data coming from a factor model, with

$$\mathbf{y}_i = \mathbf{A}\mathbf{x}_i + \boldsymbol{\epsilon}_i, \quad i = 1, \dots, n,$$

we still assume that the $\mathbf{y}_i$'s are independent of each other. Furthermore, we assuming the following.

- (F1) The series $\{\boldsymbol{\epsilon}_i\}$ has $\boldsymbol{\epsilon}_i = \Sigma_\epsilon^{1/2} \boldsymbol{\xi}_i$, where $\boldsymbol{\xi}_i$ is a $p \times 1$ vector of independent and identically distributed random variables ξ_{ij} . Each ξ_{ij} has mean 0 and unit variance, and $E|\xi_{ij}|^k \leq B < \infty$ for some constant B and $k \leq 20$. The factor series $\{\mathbf{x}_t\}$ has a constant dimension r , and $\mathbf{x}_t = \Sigma_x^{1/2} \mathbf{x}_t^*$ where $\mathbf{x}_t^*$ is a $r \times 1$ vector of independent and identically distributed random variables x_{ti}^* . Also, $E|x_{ti}^*|^k \leq B < \infty$ for some constant B and $2 < k \leq 20$.
- (F2) The covariance matrix $\Sigma_x = \text{var}(\mathbf{x}_i)$ is such that $\|\Sigma_x\| = O(1)$. The covariance matrix $\Sigma_\epsilon = \text{var}(\boldsymbol{\epsilon}_i)$ also has $\|\Sigma_\epsilon\| = O(1)$. Both matrices are non-random. The factor loading matrix $\mathbf{A}$ is such that $\|\mathbf{A}\|_F^2 = O(p)$.

Assumptions (A1) and (A2) here coincide with assumptions (A1)' and (A2)', while (A3) and (A4) coincide with (A3) and (A4) in Lam (2016). They are needed for proving the asymptotic efficiency for our estimator $\Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2)$. For data from a factor model, (F1) is assumption (F1)', while (F2) coincides with (F2) in Lam (2016). We first present some asymptotic properties of the estimated weight

$$\hat{\delta} = \frac{\text{tr}[(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})\tilde{\Sigma}_2]}{\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2}, \quad (2.6)$$

which uses $\mathbf{Y}_1$ for the construction of $\mathbf{T}$, and substitutes Σ_0 in the result of Theorem 1 by $\tilde{\Sigma}_2$.

Theorem 2. *Let Assumptions (A1)-(A2) be satisfied. Assume the split location $m = m(n)$ satisfies the constraint $\sum_{n \geq 1} p(n - m)^{-5} < \infty$ while $p = p(n)$ satisfies $p/n \rightarrow c > 0$. If $p^{-1} \text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2 \nrightarrow 0$ and is finite in probability/almost surely, then we have $\hat{\delta} - \delta \rightarrow 0$ in probability/almost surely, where δ is as in the result of Theorem 1 with $\mathbf{T}$ constructed using $\mathbf{Y}_1$ and $\tilde{\Sigma}_{\mathbf{T}}$ replaced by $\tilde{\Sigma}_{\mathbf{T}} = \mathbf{P}_1 \text{diag}(\mathbf{P}_1^T \mathbf{T} \mathbf{P}_1) \mathbf{P}_1^T$. Furthermore, supposing $\Sigma_0 \neq \sigma^2 \mathbf{I}_p$, if we have $\|\mathbf{T} - \Sigma_0\| \rightarrow 0$ in probability/almost surely and $\|\mathbf{T}\|$ is finite, then $\hat{\delta} \rightarrow 1$ and $\|\Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2) - \Sigma_0\| \rightarrow 0$ in probability/almost surely.*

For data from a factor model, let Assumptions (F1) and (F2) be satisfied. Assume the split location $m = m(n)$ satisfies the constraint $\sum_{n \geq 1} p(n - m)^{-5} < \infty$ while $p = p(n)$ satisfies $p/n \rightarrow c > 0$. If $p^{-2} \text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2 \nrightarrow 0$ and is finite in probability/almost surely, then we have $\hat{\delta} - \delta \rightarrow 0$ in probability/almost surely. Furthermore, if $p^{-1} \|\mathbf{T} - \Sigma_0\| \rightarrow 0$ in probability/almost surely and $p^{-1} \|\mathbf{T}\|$ is finite, then $\hat{\delta} \rightarrow 1$ and $p^{-1} \|\Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2) - \Sigma_0\| \rightarrow 0$ in probability/almost surely.

In words, our estimated weight approaches the true weight using data $\mathbf{Y}_1$ if $\mathbf{T}$ is significantly different from the form $\mathbf{P}_1 \mathbf{D} \mathbf{P}_1^T$ (so that $p^{-1} \text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2$ is not going to 0). If $\mathbf{T}$ approaches Σ_0 in the spectral norm, then $\mathbf{T}$ is a good enough estimator, and the theoretical result matches our intuition, that $\hat{\delta}$ should go to 1, and favor $\mathbf{T}$ completely. The resulting estimator then approaches Σ_0 in the spectral norm as well.

For data from a factor model, Σ_0 is spiked with a few eigenvalues of order p from the assumptions. Then the theorem says that if $\mathbf{T}$ can estimate those spiked eigenvalues accurate enough such that $p^{-1} \|\mathbf{T} - \Sigma_0\|$ goes to 0, we still favor $\mathbf{T}$ over a rotation-equivariant estimator completely.

These results are very useful in practice. For example, if Σ_0 is banded, then a banded estimator $\mathbf{T}$ will have $\|\mathbf{T} - \Sigma_0\|$ go to 0 in probability at a certain rate (see Bickel and Levina (2008b) for details). It means that if we are using $\Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2)$ as our estimator, then it asymptotically approaches $\mathbf{T}$ itself since $\hat{\delta}$ approaches 1, which is desirable. For data from a factor model, POET from Fan et al. (2013) can serve as $\mathbf{T}$ and our results above can be applied as well.

To present efficiency properties of our estimator, we define the ideal estimator for our purpose of comparison as

$$\Sigma_{\text{Ideal}} = \Sigma(\mathbf{P}, \mathbf{T}, \Sigma_0),$$

which is the same as the one in (2.3) except that $\mathbf{T}$ here is defined as the one constructed using $\mathbf{Y}_1$. The efficiency loss of an estimator $\hat{\Sigma}$ can then be defined as

$$EL(\Sigma_0, \hat{\Sigma}) = 1 - \frac{\|\Sigma_{\text{Ideal}} - \Sigma_0\|_F^2}{\|\hat{\Sigma} - \Sigma_0\|_F^2}. \quad (2.7)$$

When $\hat{\Sigma}$ has a larger Frobenius loss than the ideal estimator, the efficiency loss is positive, and is negative vice versa.

Theorem 3. *Let Assumptions (A1)-(A4) be satisfied, and $\Sigma_0 \neq \sigma^2 \mathbf{I}_p$. Assume the split location $m = m(n)$ satisfies the constraints $m/n \rightarrow 1, n - m \rightarrow \infty$ and $\sum_{n \geq 1} p(n - m)^{-5} < \infty$ while $p = p(n)$ satisfies $p/n \rightarrow c > 0$. If $p^{-1} \text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2, p^{-1} \text{tr}(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})^2 \nrightarrow 0$ and both, together with $\|\mathbf{T}\|$, are finite in probability/almost surely, then $EL(\Sigma_0, \Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2)) \rightarrow 0$ in probability/almost surely.*

We do not attempt to prove asymptotic efficiency when the data is from a factor model in this paper. We illustrate the corresponding results in our simulations instead, when a regularized covariance matrix

estimator like POET from Fan et al. (2013) is used as $\mathbf{T}$ in (2.5). Note that $\mathbf{\Sigma}_0 \neq \sigma^2 \mathbf{I}_p$ in the theorem. As discussed earlier, the problem actually reduces to the one in Lam (2016) when $\mathbf{\Sigma}_0 = \sigma^2 \mathbf{I}_p$, and our estimator reduces to NERCOME too in Lam (2016), which still works well empirically (see the simulation results in Lam (2016) for more details).

3 Extension to Two Regularized Matrices

When we are truly uncertain what assumptions to make on $\mathbf{\Sigma}_0$, it is natural for us to try more than one regularization methods. In this section we introduce two regularized estimators to be included in a linear combination with a rotation-equivariant estimator. This way, we do not need to determine what regularization methods are more appropriate in advance if, like the results in Theorem 2, the weights are going to 0 or 1 matching our intuition when a particular regularization is more appropriate than the other.

Given two regularized estimators $\mathbf{T}_1$ and $\mathbf{T}_2$ and a fixed orthogonal matrix $\mathbf{P}$ (we use the eigenmatrix from the eigen-decomposition of the sample covariance matrix), we consider

$$\min_{\delta_1, \delta_2, \mathbf{D}} \left\| (1 - \delta_1 - \delta_2) \mathbf{P} \mathbf{D} \mathbf{P}^T + \delta_1 \mathbf{T}_1 + \delta_2 \mathbf{T}_2 - \mathbf{\Sigma}_0 \right\|_F^2. \quad (3.1)$$

Theorem 4. Suppose $\delta_1 + \delta_2 \neq 1$, and $\mathbf{T}_1$, $\mathbf{T}_2$ and $\mathbf{T}_1 - \mathbf{T}_2$ are not of the form $\mathbf{P} \mathbf{D} \mathbf{P}^T$ for some diagonal matrix $\mathbf{D}$. Then defining $\hat{\mathbf{\Sigma}}_{\mathbf{T}_i} = \mathbf{P} \text{diag}(\mathbf{P}^T \mathbf{T}_i \mathbf{P}) \mathbf{P}^T$, the solution to the minimization problem (3.1) is

$$\begin{aligned} \delta_1 &= \frac{\text{tr}[(\mathbf{T}_1 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_1}) \mathbf{\Sigma}_0] \text{tr}(\mathbf{T}_2 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_2})^2 - \text{tr}[(\mathbf{T}_2 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_2})(\mathbf{T}_1 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_1]) \text{tr}[(\mathbf{T}_2 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_2}) \mathbf{\Sigma}_0]}{\text{tr}(\mathbf{T}_2 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_2})^2 \text{tr}(\mathbf{T}_1 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_1})^2 - \text{tr}^2[(\mathbf{T}_2 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_2})(\mathbf{T}_1 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_1})]}, \\ \delta_2 &= \frac{\text{tr}[(\mathbf{T}_2 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_2}) \mathbf{\Sigma}_0] \text{tr}(\mathbf{T}_1 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_1})^2 - \text{tr}[(\mathbf{T}_2 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_2})(\mathbf{T}_1 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_1]) \text{tr}[(\mathbf{T}_1 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_1}) \mathbf{\Sigma}_0]}{\text{tr}(\mathbf{T}_2 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_2})^2 \text{tr}(\mathbf{T}_1 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_1})^2 - \text{tr}^2[(\mathbf{T}_2 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_2})(\mathbf{T}_1 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_1})]}, \\ \mathbf{D} &= \frac{1}{1 - \delta_1 - \delta_2} (\text{diag}(\mathbf{P}^T \mathbf{\Sigma}_0 \mathbf{P}) - \delta_1 \text{diag}(\mathbf{P}^T \mathbf{T}_1 \mathbf{P}) - \delta_2 \text{diag}(\mathbf{P}^T \mathbf{T}_2 \mathbf{P})). \end{aligned}$$

The regularity conditions are sound, since the violation of one or more of these reduces the problem to either (2.2) or even the one considered in Lam (2016). Same thing happens when $\mathbf{\Sigma}_0 = \sigma^2 \mathbf{I}_p$, since then $\delta_1 = \delta_2 = 0$. And, the denominator in δ_1 and δ_2 is 0 when any one of the regularity conditions are violated. The resulting estimator is given by

$$\mathbf{\Sigma}(\mathbf{P}, \mathbf{T}, \mathbf{\Sigma}_0) = \mathbf{P} \text{diag}(\mathbf{P}^T \mathbf{\Sigma}_0 \mathbf{P}) \mathbf{P}^T + \delta_1 (\mathbf{T}_1 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_1}) + \delta_2 (\mathbf{T}_2 - \hat{\mathbf{\Sigma}}_{\mathbf{T}_2}). \quad (3.2)$$

3.1 Proposed estimator and theoretical results

We use the same data splitting idea as in Section 2.2, and consider the minimization problem

$$\min_{\delta_1, \delta_2, \mathbf{D}} \left\| (1 - \delta_1 - \delta_2) \mathbf{P}_1 \mathbf{D} \mathbf{P}_1^T + \delta_1 \mathbf{T}_1 + \delta_2 \mathbf{T}_2 - \mathbf{\Sigma}_0 \right\|_F^2,$$

where $\mathbf{T}_1$ and $\mathbf{T}_2$ are both constructed from the data matrix $\mathbf{Y}_1$. Substituting $\mathbf{\Sigma}_0$ by $\tilde{\mathbf{\Sigma}}_2$ in the resulting solution, we have our estimator

$$\mathbf{\Sigma}(\mathbf{P}_1, \mathbf{T}, \tilde{\mathbf{\Sigma}}_2) = \mathbf{P}_1 \text{diag}(\mathbf{P}_1^T \tilde{\mathbf{\Sigma}}_2 \mathbf{P}_1) \mathbf{P}_1^T + \hat{\delta}_1 (\mathbf{T}_1 - \tilde{\mathbf{\Sigma}}_{\mathbf{T}_1}) + \hat{\delta}_2 (\mathbf{T}_2 - \tilde{\mathbf{\Sigma}}_{\mathbf{T}_2}), \quad (3.3)$$

where $\hat{\delta}_i$ is the same as the corresponding δ_i in the results of Theorem 4, except that Σ_0 is substituted by $\tilde{\Sigma}_2$ and $\hat{\Sigma}_{\mathbf{T}_i}$ is replaced by $\tilde{\Sigma}_{\mathbf{T}_i} = \mathbf{P}_1 \text{diag}(\mathbf{P}_1^\top \mathbf{T}_i \mathbf{P}_1) \mathbf{P}_1^\top$.

To present further results, define $a_{ij} = \text{tr}[(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})(\mathbf{T}_j - \tilde{\Sigma}_{\mathbf{T}_j})]$ for $i, j = 1, 2$.

Theorem 5. *Let Assumptions (A1)-(A2) be satisfied. Assume the split location $m = m(n)$ satisfies the constraint $\sum_{n \geq 1} p(n - m)^{-5} < \infty$ while $p = p(n)$ satisfies $p/n \rightarrow c > 0$. If $p^{-1}a_{ii}, p^{-2}(a_{11}a_{22} - a_{12}^2) \rightarrow 0$ and are finite in probability/almost surely, then we have $\hat{\delta}_i - \delta_i \rightarrow 0$ in probability/almost surely for $i = 1, 2$, where δ_i is as in the result of Theorem 4 with $\mathbf{T}_i$ constructed using $\mathbf{Y}_1$ and $\tilde{\Sigma}_{\mathbf{T}_i} = \mathbf{P}_1 \text{diag}(\mathbf{P}_1^\top \mathbf{T}_i \mathbf{P}_1) \mathbf{P}_1^\top$. Furthermore, supposing $\Sigma_0 \neq \sigma^2 \mathbf{I}_p$, if we have $\|\mathbf{T}_i - \Sigma_0\| \rightarrow 0$ in probability/almost surely while $\|\mathbf{T}_{3-i} - \Sigma_0\| \rightarrow 0$ but is finite for $i = 1, 2$, and $\|\mathbf{T}_1\|, \|\mathbf{T}_2\|$ are finite as well, then $\hat{\delta}_i \rightarrow 1$, $\hat{\delta}_{3-i} \rightarrow 0$ and $\|\Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2) - \Sigma_0\| \rightarrow 0$ in probability/almost surely.*

For data from a factor model, let Assumptions (F1) and (F2) be satisfied. Assume the split location $m = m(n)$ satisfies the constraint $\sum_{n \geq 1} p(n - m)^{-5} < \infty$ while $p = p(n)$ satisfies $p/n \rightarrow c > 0$. If $p^{-2}a_{ii}, p^{-4}(a_{11}a_{22} - a_{12}^2) \rightarrow 0$ and are finite in probability/almost surely, then we have $\hat{\delta}_i - \delta_i \rightarrow 0$ in probability/almost surely for $i = 1, 2$. Furthermore, if $p^{-1}\|\mathbf{T}_i - \Sigma_0\| \rightarrow 0$ in probability/almost surely while $p^{-1}\|\mathbf{T}_{3-i} - \Sigma_0\| \rightarrow 0$ but is finite for $i = 1, 2$, and $p^{-1}\|\mathbf{T}_1\|, p^{-1}\|\mathbf{T}_2\|$ are finite as well, then $\hat{\delta}_i \rightarrow 1$, $\hat{\delta}_{3-i} \rightarrow 0$ and $p^{-1}\|\Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2) - \Sigma_0\| \rightarrow 0$ in probability/almost surely.

Theorem 5 tells us that our estimator is adaptive to the inclusion of two regularized estimators, and favors the correct one asymptotically. This is a very useful property when we want to take advantages of different regularization methods. See the simulation results in Section 5 for more details.

Similar to Section 2.3, we define the ideal estimator as $\Sigma_{\text{Ideal}} = \Sigma(\mathbf{P}, \mathbf{T}, \Sigma_0)$, which is the one in (3.2) except that $\mathbf{T}_1$ and $\mathbf{T}_2$ are constructed using $\mathbf{Y}_1$. Define b_{ij} to be the a_{ij} defined above except that $\tilde{\Sigma}_{\mathbf{T}_i}$ is replaced by $\hat{\Sigma}_{\mathbf{T}_i} = \mathbf{P} \text{diag}(\mathbf{P}^\top \mathbf{T}_i \mathbf{P}) \mathbf{P}^\top$.

Theorem 6. *Let Assumptions (A1)-(A4) be satisfied, and $\Sigma_0 \neq \sigma^2 \mathbf{I}_p$. Assume the split location $m = m(n)$ satisfies the constraints $m/n \rightarrow 1, n - m \rightarrow \infty$ and $\sum_{n \geq 1} p(n - m)^{-5} < \infty$ while $p = p(n)$ satisfies $p/n \rightarrow c > 0$. If $p^{-1}a_{ii}, p^{-1}\hat{a}_{ii}, p^{-2}(a_{11}a_{22} - a_{12}^2), p^{-2}(b_{11}b_{22} - b_{12}^2) \rightarrow 0$ and all of these, together with $\|\mathbf{T}\|$, are finite in probability/almost surely, then $EL(\Sigma_0, \Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2)) \rightarrow 0$ in probability/almost surely, where $EL(\Sigma_0, \hat{\Sigma})$ is as define in (2.7) using $\Sigma_{\text{Ideal}} = \Sigma(\mathbf{P}, \mathbf{T}, \Sigma_0)$.*

4 Properties of an Averaged Estimator

In this section, we illustrate an improved estimator using averaging when two regularized covariance matrices are concerned. The same idea can be applied to the version with one regularized estimator. Since the covariance matrices in (2.5) and (3.3) are both constructed from the data $\mathbf{Y}_1$, and the $\mathbf{y}_i$'s are all independent of each other, a new permutation of the data will result in a different $\mathbf{Y}_1$ even if we use the same split location m . Similar to Lam (2016), we permute the data each time, and for the j th permutation where $j = 1, \dots, M$, we use the resulting data $\mathbf{Y}^{(j)} = (\mathbf{Y}_1^{(j)}, \mathbf{Y}_2^{(j)})$ to construct

$$\hat{\Sigma}_m^{(j)} = \Sigma(\mathbf{P}_{1j}, \mathbf{T}_j, \tilde{\Sigma}_2^{(j)}) = \mathbf{P}_{1j} \text{diag}(\mathbf{P}_{1j}^\top \tilde{\Sigma}_2^{(j)} \mathbf{P}_{1j}) \mathbf{P}_{1j}^\top + \hat{\delta}_{1j}(\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}}) + \hat{\delta}_{2j}(\mathbf{T}_{2j} - \tilde{\Sigma}_{\mathbf{T}_{2j}}). \quad (4.1)$$

In the above, $\tilde{\Sigma}_1^{(j)} = m^{-1} \mathbf{Y}_1^{(j)} \mathbf{Y}_1^{(j)\top} = \mathbf{P}_{1j} \mathbf{D}_{1j} \mathbf{P}_{1j}^\top$, and $\tilde{\Sigma}_2^{(j)} = (n-m)^{-1} \mathbf{Y}_2^{(j)} \mathbf{Y}_2^{(j)\top}$. Each $\mathbf{T}_{ij}$ denotes the i th regularized estimator constructed from $\mathbf{Y}_1^{(j)}$ in the j th permutation, $i = 1, 2$, $j = 1, \dots, M$. Finally, for $i = 1, 2$ and $j = 1, \dots, M$, $\hat{\delta}_{ij}$ is the same as the corresponding δ_i in the result of Theorem 4, except that Σ_0 is substituted by $\tilde{\Sigma}_2^{(j)}$, and $\mathbf{T}_i$ by $\mathbf{T}_{ij}$ with $\tilde{\Sigma}_{\mathbf{T}_{ij}} = \mathbf{P}_{1j} \text{diag}(\mathbf{P}_{1j}^\top \mathbf{T}_{ij} \mathbf{P}_{1j}) \mathbf{P}_{1j}^\top$. We can then average all these estimators and arrive at the final estimator

$$\hat{\Sigma}_{m,M} = \frac{1}{M} \sum_{j=1}^M \hat{\Sigma}_m^{(j)}. \quad (4.2)$$

In practice, using $M = 50$ achieves a good trade-off between accuracy and computational efficiency. The resulting estimator is usually much better than using just $M = 1$ alone.

Theorem 7. *Let Assumptions (A1)-(A4) be satisfied, and $\Sigma_0 \neq \sigma^2 \mathbf{I}_p$. Assume the split location $m = m(n)$ satisfies the constraints $m/n \rightarrow 1$, $n-m \rightarrow \infty$ and $\sum_{n \geq 1} p(n-m)^{-5} < \infty$ while $p = p(n)$ satisfies $p/n \rightarrow c > 0$. If the corresponding $p^{-1}a_{ii}$, $p^{-1}b_{ii}$, $p^{-2}(a_{11}a_{22} - a_{12}^2)$, $p^{-2}(b_{11}b_{22} - b_{12}^2) \rightarrow 0$, and together with $\|\mathbf{T}_i\|$, are all finite in probability/almost surely for all M permutations, then $EL(\Sigma_0, \hat{\Sigma}_{m,M}) \leq 0$ in probability/almost surely.*

Furthermore, if $\sum_{n \geq 1} (n-m)^{-3} < \infty$, then in probability/almost surely, we have

$$\frac{1}{p}(\text{tr}(\hat{\Sigma}_{m,M}) - \text{tr}(\Sigma_0)) \rightarrow 0, \quad \frac{1}{p}\text{tr}(\hat{\Sigma}_{m,M}\Sigma_0) \geq \lambda_{\min}^2(\Sigma_0),$$

where $\lambda_{\min}(\mathbf{A})$ denotes the minimum eigenvalue of a matrix $\mathbf{A}$. The first trace property above are true if the data is from a factor model with Assumptions (A1),(A2) replaced by (F1),(F2), and the corresponding $p^{-2}a_{ii}$, $p^{-4}(a_{11}a_{22} - a_{12}^2) \rightarrow 0$ and are finite in probability/almost surely for all M permutations. The second trace property is true if it is modified to

$$\frac{1}{p^3}\text{tr}(\hat{\Sigma}_{m,M}\Sigma_0) \geq \frac{1}{p^2}\lambda_{\min}^2(\Sigma_0).$$

The above theorem shows that asymptotic efficiency still holds for the averaged estimator. Moreover, although we cannot prove the asymptotic positive definiteness of the estimator $\hat{\Sigma}_{m,M}$ in general, the two trace properties are important characteristics for our estimator. This is because the true covariance matrix Σ_0 also satisfies the two trace properties above.

4.1 Speed boosting and choice of split location

The estimator $\hat{\Sigma}_{m,M}$ in (4.2) involves calculating the regularized covariance matrices $\mathbf{T}_1$ and $\mathbf{T}_2$ M times for a particular split location m . If we search for several split locations (see below), then the computational burden is high even when p is not too large.

To speed up the calculation of the estimator, we concentrate on banding in Bickel and Levina (2008b) and POET in Fan et al. (2013) for $\mathbf{T}_1$ and $\mathbf{T}_2$. POET includes generalized thresholding when set with $K = 0$ factor. To speed up the repeated calculations of the banding estimators, we first find the banding number k chosen by a 5-fold cross-validation (see Bickel and Levina (2008b) for details) when the whole data set $\mathbf{Y}$ is used. Then we use such a k to band each $\mathbf{T}_{1j}$ constructed from the permuted data set $\mathbf{Y}^{(j)}$.

For speeding up the repeated calculations of the POET estimators with K factors, note that an eigen-decomposition of the sample covariance matrix $\tilde{\Sigma}_1^{(j)}$ is needed in both the calculation of $\mathbf{P}_{1j}$ and the factor loading matrix in the POET method, and hence one eigen-decomposition is all that is needed in each repeated calculation. It means that we are using the solution $p^{1/2}$ times the matrix of the K column vectors in $\mathbf{P}_{1j}$ corresponding to the K largest eigenvalues of $\tilde{\Sigma}_1^{(j)}$ to be the estimated factor loading matrix (see for example Bai and Ng (2002) for more details). We indeed implement this in our codes to minimize the number of eigen-decompositions required.

One major parameter we need to decide is the split location for our estimator. From Theorems 3, 6 and 7, the conditions that have to be satisfied for the split location are that $\sum_{n \geq 1} p(n-m)^{-5} < \infty$, $n-m \rightarrow \infty$ and $m/n \rightarrow 1$. Coupling with the high dimensional assumption that $p/n \rightarrow c > 0$, one possible solution is to set $m = n - an^{1/2}$. This is indeed a solution adopted by Lam (2016), and we propose a similar criterion for choosing m here:

$$g(m) = \left\| \hat{\Sigma}_{m,M} - \frac{1}{M} \sum_{j=1}^M \tilde{\Sigma}_2^{(j)} \right\|_F^2, \quad (4.1)$$

where $\tilde{\Sigma}_2^{(j)}$ is defined in (4.1). We choose m that minimizes $g(m)$ above. In practice we search the following 7 split locations for minimizing $g(m)$:

$$m = [2n^{1/2}, 0.2n, 0.4n, 0.6n, 0.8n, n - 2.5n^{1/2}, n - 1.5n^{1/2}].$$

The last two split locations are of the form $n - an^{1/2}$. The four locations $0.2n$ to $0.8n$ are there to accommodate finite sample performance. The split location $2n^{1/2}$ is for the case $\Sigma_0 = \sigma^2 \mathbf{I}_p$, when all proposed estimators are reduced to NERCOME in Lam (2016) asymptotically, and it is discussed in Lam (2016) explicitly that this case requires m to be as small as possible while still going to infinity. The order $n^{1/2}$ is a good choice in the end.

4.2 Other practical concerns

The regularity condition that $\mathbf{T}$ cannot be of the form $\mathbf{PDP}^T$ in Theorem 1 does have a bearing in practice. For instance, when $\mathbf{T}$ is a banded or a thresholded estimator, it can be that $\mathbf{T}$ becomes a diagonal matrix and somehow becomes “too close” to the form $\mathbf{PDP}^T$ because the diagonal elements in $\mathbf{T}$ are very similar. This will make $\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2$ much smaller than usual, and hence δ will be inflated in magnitude, making the estimator $\Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2)$ in (2.5) unstable. The same holds true when integrating two regularized estimators $\mathbf{T}_1$ and $\mathbf{T}_2$. For instance if $\mathbf{T}_1$ is banded and $\mathbf{T}_2$ is soft-thresholded, they can both be diagonal with almost identical diagonal elements. This will make the denominator in δ_1 and δ_2 in Theorem 4 very small, making the estimator $\Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2)$ in (3.3) unstable.

In practice, we find that when a weight has a magnitude larger than 3 (can be δ or $1 - \delta$, or δ_1, δ_2 or $1 - \delta_1 - \delta_2$), the corresponding estimator exhibits substantial instability. For a single $\mathbf{T}$, it means that $\mathbf{T}$ is starting to be too close to the form $\mathbf{PDP}^T$, so that the whole problem is approaching the construction of NERCOME in Lam (2016). Hence we impose our estimator to be exactly the NERCOME estimator whenever this happens to protect it from becoming too unstable. For the averaged estimator $\hat{\Sigma}_{m,M}$ in (4.2), we monitor how many times the estimator is made exactly the NERCOME among the M permutations.

Practically, for more accurate estimator, we want at least 30 of the estimators among the M permutations are not made exactly to NERCOME. Suppose there are $M_e \geq 30$ permutations, indexed by the set S so that $M_e = |S|$, where the estimator is not made to NERCOME exactly. Then our average estimator becomes

$$\hat{\Sigma}_{m,M} = \frac{1}{M} \sum_{j=1}^M \mathbf{P}_{1j} \text{diag}(\mathbf{P}_{1j}^T \tilde{\Sigma}_2^{(j)} \mathbf{P}_{1j}) \mathbf{P}_{1j}^T + \frac{1}{M_e} \sum_{j \in S} \frac{\text{tr}[(\mathbf{T}_j - \tilde{\Sigma}_{\mathbf{T}_j}) \tilde{\Sigma}_2^{(j)}]}{\text{tr}(\mathbf{T}_j - \tilde{\Sigma}_{\mathbf{T}_j})^2} (\mathbf{T}_j - \tilde{\Sigma}_{\mathbf{T}_j}). \quad (4.2)$$

If $M_e < 30$, we just take the first term in (4.2) above, which is exactly the averaged NERCOME proposed in Lam (2016). This works very well in practice and reduces instability substantially.

We use similar rules for our averaged estimator for integrating $\mathbf{T}_1$ and $\mathbf{T}_2$. For a particular permutation, if any of $\hat{\delta}_1, \hat{\delta}_2$ or $1 - \hat{\delta}_1 - \hat{\delta}_2$ are larger than 3 in magnitude, then we discard it. Defining the set S as above, with $M_e = |S|$, if $M_e \geq 30$, the resulting averaged estimator is then

$$\hat{\Sigma}_{m,M} = \frac{1}{M} \sum_{j=1}^M \mathbf{P}_{1j} \text{diag}(\mathbf{P}_{1j}^T \tilde{\Sigma}_2^{(j)} \mathbf{P}_{1j}) \mathbf{P}_{1j}^T + \frac{1}{M_e} \sum_{j \in S} \left(\hat{\delta}_{1j} (\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}}) + \hat{\delta}_{2j} (\mathbf{T}_{2j} - \tilde{\Sigma}_{\mathbf{T}_{2j}}) \right). \quad (4.3)$$

If $M_e < 30$, we reduce the above to the one in (4.2), getting two averaged estimators, one for $\mathbf{T}_1$ and one for $\mathbf{T}_2$. If say $\mathbf{T}_1$ has $M_e \geq 30$ and $\mathbf{T}_2$ has $M_e < 30$, then we choose the averaged estimator with $\mathbf{T}_1$ and vice versa. If $M_e \geq 30$ for both $\mathbf{T}_1$ and $\mathbf{T}_2$, then it means that originally $\mathbf{T}_1$ and $\mathbf{T}_2$ are perhaps too similar, causing instability in the final estimator. Hence we average the two integrated estimators with $\mathbf{T}_1$ and $\mathbf{T}_2$ in this case. Finally, if both $M_e < 30$, then it is reduced to the averaged NERCOME estimator. We impose all these rules in our simulations and real data analysis.

5 Empirical Results

5.1 Simulation Experiments

We have three proposed estimators to be compared with other methods in this section. They are the estimator integrated with Banding (abbreviated as INT-BAND), the one integrated with POET (abbreviated as INT-POET), and finally the one integrated with both (abbreviated as INT-Double). For POET and related methods, we use $C = 0.5$ throughout. We use the averaged estimator as in (4.2), and to speed up the simulations, we use $M = 40$ and search for 4 split locations

$$m = [2n^{1/2}, 0.3n, 0.7n, n - 2n^{1/2}].$$

This way, the time to compute our estimator is cut by half, without losing accuracy and efficiency practically. Other methods to be compared with our estimators include Banding (with banding number chosen by 5-fold cross-validation for each simulated data set) and POET themselves, with POET includes pure adaptive thresholding as a special case. We also compare to three eigenvalues shrinkage methods, namely, the Nonparametric Eigenvalues-Regularized COvariance Matrix Estimator (NERCOME) from Lam (2016), the nonlinear shrinkage estimator (NONLIN) from Ledoit and Wolf (2012), and the grand average estimator (Grand Avg) from Abadir et al. (2014). Finally, we also compare to the NOVELIST estimator from

Huang and Fryzlewicz (2015), which combines the sample covariance matrix with a thresholded estimator. We create 10 different profiles below.

- (I) Sparse Σ_0 with 20% non-zeros, generated randomly each time, $\mathbf{y}_t \sim N(\mathbf{0}, \Sigma_0)$.
- (II) General non-sparse $\Sigma_0 = \mathbf{Q}\mathbf{D}\mathbf{Q}^T$, where $\mathbf{Q}$ is an orthogonal matrix generated randomly each time, and $\mathbf{D}$ is diagonal with values 1, 4, 7 and 10 each appears 25% of times, $\mathbf{y}_t \sim N(\mathbf{0}, \Sigma_0)$.
- (III) (Sparse + Banded) $\Sigma_0 = \Sigma_1 + \Sigma_2$, where Σ_1 is the same as the Σ_0 in profile (I), and $\Sigma_2 = (0.9^{|i-j|})_{1 \leq i, j \leq p}$, $\mathbf{y}_t \sim N(\mathbf{0}, \Sigma_0)$.
- (IV) (Cross) $\Sigma_0 = \mathbf{Q}\text{diag}(\lambda_1 \mathbf{1}_{\{\lambda_1 > 0\}}, \dots, \lambda_p \mathbf{1}_{\{\lambda_p > 0\}})\mathbf{Q}^T + \mathbf{I}_p$, where $\mathbf{Q}\text{diag}(\lambda_1, \dots, \lambda_p)\mathbf{Q}^T$ is the eigen-decomposition of the left-right flipped matrix of $(0.9^{|i-j|})_{1 \leq i, j \leq p}$, $\mathbf{y}_t \sim N(\mathbf{0}, \Sigma_0)$.
- (V) (Factor model) $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{E}$, where $\mathbf{A}$ is $p \times 2$, $\mathbf{X} = (x_1, \dots, x_n)$ is $2 \times n$, and finally $\mathbf{E} = \Sigma^{1/2}\mathbf{Z}$, where Σ is the Σ_0 in profile (I). The factor loading matrix $\mathbf{A}$, the matrix of factor series $\mathbf{X}$ and $\mathbf{Z}$ are generated each time with independent and identically distributed $N(0, 1)$ elements.
- (It-IVt) Same as the corresponding profiles (I) to (IV), except that $\mathbf{Y} = \Sigma_0^{1/2}\mathbf{Z}$ with $\mathbf{Z}$ having independent and identically distributed t_5 elements.
- (Vt) Same as the corresponding profile (V), except that $\mathbf{E} = \Sigma^{1/2}\mathbf{Z}$ where $\mathbf{Z}$ has independent and identically distributed t_5 elements.

Profile (I) favors POET ($K = 0$), while profile (III) favors banding and profile (V) POET with $K = 2$. Profile (IV) also has sparse Σ_0 , but the cross-shaped non-zero pattern on Σ_0 means it is partly banded as well. Thresholding or banding alone are not able to take full advantage of the structure of Σ_0 then. The 5 other profiles are created to test the robustness of our methods to fat-tailed distribution.

We simulate 500 times from the 10 profiles under $n = 200$ and $p = 100, 200, 400$. We use the following 7 loss functions for comparisons:

$$\begin{aligned}
L_1(\Sigma, \hat{\Sigma}) &= \|\Sigma - \hat{\Sigma}\|_F, & L_2(\Sigma, \hat{\Sigma}) &= \text{tr}(\Sigma \hat{\Sigma}^{-1}) - \log \det(\Sigma \hat{\Sigma}^{-1}) - p, \\
L_3(\Sigma, \hat{\Sigma}) &= \|\Sigma^{-1} - \hat{\Sigma}^{-1}\|, & L_4(\Sigma, \hat{\Sigma}) &= \text{tr}(\Sigma^{-1} \hat{\Sigma}) - \log \det(\Sigma^{-1} \hat{\Sigma}) - p, \\
L_5(\Sigma, \hat{\Sigma}) &= \|\Sigma - \hat{\Sigma}\|, & L_6(\Sigma, \hat{\Sigma}) &= \sum_{i=1}^p \lambda_i^{1/2}(\Sigma - \hat{\Sigma})^2, & L_7(\Sigma, \hat{\Sigma}) &= \text{tr}(\Sigma + \hat{\Sigma} - 2\Sigma^{1/2}\hat{\Sigma}^{1/2}).
\end{aligned}$$

The L_1 loss is the Frobenius loss which our estimator is supposed to minimize. The second is the inverse Stein's loss, and the fourth one is the Stein's loss. We also include L_5 the spectral loss and L_6 the sum of singular values of $\Sigma - \hat{\Sigma}$, while L_7 is called the Fréchet loss. All of them are non-negative and is 0 when $\hat{\Sigma} = \Sigma$.

From Table 1, under profile (I), the INT-BAND and NERCOME have virtually the same performance. This is understandable, since Banding is performing badly, and hence the δ in INT-BAND is small, virtually turning it into NERCOME. POET, being the adaptive soft-thresholding with $K = 0$, is doing well since Σ_0 is indeed sparse in profile (I). It is therefore remarkable for INT-POET, which is the combination of NERCOME and POET, to outperform both NERCOME and POET in all 7 losses. Even more remarkable

$n = 200$		Profile (I)						
$p = 100$	$\delta(\%)$	L_1	L_2	L_3	L_4	L_5	L_6	L_7
INT-BAND	-4.7 _(9.6)	37.1 _(1.1)	10.1 _(0.7)	0.8 _(0.1)	15.6 _(2.0)	8.3 _(0.5)	165.4 _(2.5)	39.3 _(1.5)
Banding	-	50.2 _(5.7)	17.2 _(6.8)	1.0 _(0.1)	30.7 _(4.1)	10.7 _(1.9)	190.9 _(9.9)	70.4 _(10.1)
INT-POET	44.7 _(6.9)	35.2 _(1.2)	9.2 _(0.6)	0.8 _(0.1)	14.3 _(1.9)	7.9 _(0.4)	161.0 _(2.8)	35.4 _(1.6)
POET	-	37.6 _(1.5)	10.2 _(0.6)	0.9 _(0.0)	14.2 _(1.3)	8.0 _(0.4)	166.3 _(3.4)	38.9 _(1.6)
INT-Double	$\delta_1 : -17.2_{(20.8)}$ $\delta_2 : 47.3_{(7.7)}$	35.0_(1.3)	9.1_(0.6)	0.8 _(0.1)	14.2_(1.9)	7.8_(0.4)	160.6_(2.9)	35.1_(1.6)
NERCOME	-	37.2 _(1.1)	10.1 _(0.7)	0.8 _(0.1)	15.6 _(2.0)	8.3 _(0.5)	165.5 _(2.5)	39.3 _(1.5)
NONLIN	-	36.6 _(1.1)	9.8 _(0.7)	0.7_(0.1)	14.2 _(1.2)	8.1 _(0.4)	165.0 _(2.4)	38.1 _(1.2)
Grand Avg	-	37.7 _(1.1)	10.8 _(0.8)	1.0 _(0.0)	20.1 _(2.0)	8.2 _(0.4)	166.6 _(2.4)	42.5 _(1.5)
NOVELIST	-	36.9 _(1.8)	10.3 _(0.8)	0.9 _(0.0)	17.9 _(2.4)	7.9 _(0.5)	165.2 _(4.1)	40.0 _(3.1)
$p = 200$	$\delta(\%)$	L_1	L_2	L_3	L_4	L_5	L_6	L_7
INT-BAND	-4.1 _(6.6)	81.8 _(1.1)	24.3 _(1.3)	0.69 _(0.02)	44.4 _(3.8)	12.5 _(0.4)	413.4 _(2.8)	140.4 _(3.5)
Banding	-	97.9 _(8.6)	36.5 _(47.0)	0.92 _(3.42)	60.1 _(6.3)	14.3 _(1.5)	451.2 _(19.1)	198.8 _(33.9)
INT-POET	22.7 _(4.6)	80.2 _(1.2)	23.5 _(1.2)	0.69 _(0.02)	43.0 _(3.7)	12.2 _(0.4)	409.5 _(3.0)	135.4 _(3.5)
POET	-	92.0 _(1.9)	30.5 _(1.1)	0.64_(0.02)	39.5_(2.5)	13.6 _(0.5)	438.6 _(4.5)	163.9 _(3.5)
INT-Double	$\delta_1 : -12.9_{(12.4)}$ $\delta_2 : 24.2_{(4.8)}$	80.0_(1.2)	23.5_(1.2)	0.69 _(0.02)	42.8 _(3.7)	12.2_(0.4)	409.1_(3.1)	134.9_(3.5)
NERCOME	-	81.8 _(1.1)	24.3 _(1.3)	0.69 _(0.02)	44.4 _(3.8)	12.5 _(0.4)	413.4 _(2.8)	140.4 _(3.5)
NONLIN	-	81.0 _(1.1)	24.0 _(1.5)	0.69 _(0.02)	43.4 _(3.4)	12.2 _(0.4)	413.0 _(2.8)	138.3 _(3.5)
Grand Avg	-	82.5 _(1.1)	24.7 _(1.3)	0.70 _(0.02)	48.1 _(4.2)	12.3 _(0.3)	414.5 _(2.8)	145.0 _(3.9)
NOVELIST	-	83.3 _(1.6)	38.6 _(2.4)	0.72 _(0.02)	45.9 _(4.0)	12.4 _(0.4)	418.6 _(3.9)	145.8 _(4.7)
$p = 200$		Profile (It)						
INT-BAND	-4.6 _(6.2)	81.8 _(1.1)	24.4 _(1.2)	0.69 _(0.02)	44.6 _(4.0)	12.5 _(0.4)	413.5 _(2.9)	140.6 _(3.5)
Banding	-	113.9 _(24.1)	38.1 _(10.8)	1.03 _(0.02)	70.0 _(17.7)	21.1 _(9.1)	481.1 _(43.2)	239.7 _(64.3)
INT-POET	20.0 _(4.7)	80.6 _(1.1)	23.8 _(1.1)	0.69 _(0.02)	43.4 _(3.9)	12.3 _(0.4)	410.5 _(2.9)	136.7 _(3.6)
POET	-	95.2 _(3.3)	31.7 _(1.0)	0.65_(0.02)	41.2_(2.5)	17.6 _(6.5)	443.3 _(5.0)	172.3 _(5.7)
INT-Double	$\delta_1 : -13.9_{(13.1)}$ $\delta_2 : 24.3_{(6.4)}$	80.1_(1.2)	23.6_(1.1)	0.69 _(0.02)	42.9 _(4.0)	12.2_(0.4)	409.3_(3.3)	135.1_(3.9)
NERCOME	-	81.8 _(1.1)	24.4 _(1.2)	0.69 _(0.02)	44.6 _(4.0)	12.5 _(0.4)	413.6 _(2.8)	140.8 _(3.6)
NONLIN	-	81.5 _(1.8)	24.4 _(1.8)	0.69 _(0.06)	43.2 _(3.2)	13.1 _(4.3)	414.1 _(3.1)	139.3 _(3.7)
Grand Avg	-	82.5 _(1.1)	24.8 _(1.2)	0.70 _(0.02)	48.6 _(4.0)	12.3 _(0.4)	414.7 _(2.8)	145.6 _(3.7)
NOVELIST	-	88.8 _(3.3)	40.5 _(2.2)	0.72 _(0.02)	48.1 _(4.1)	16.7 _(6.9)	429.0 _(4.6)	159.9 _(6.4)
$p = 400$		Profile (I)						
INT-BAND	-2.5 _(4.4)	171.2 _(1.1)	52.1 _(2.0)	0.50 _(0.01)	99.0 _(6.7)	17.9 _(0.3)	1007.6 _(3.5)	438.9 _(8.5)
Banding	-	192.2 _(17.3)	65.0 _(19.8)	0.50 _(0.01)	117.4 _(12.0)	19.7 _(2.2)	1065.7 _(45.1)	546.5 _(97.7)
INT-POET	8.5 _(1.5)	170.5 _(1.1)	51.7 _(2.0)	0.50 _(0.01)	98.2 _(6.6)	17.8 _(0.3)	1005.4 _(3.5)	435.3 _(8.3)
POET	-	221.5 _(2.8)	96.2 _(2.3)	0.47_(0.01)	107.7 _(4.6)	24.9 _(0.8)	1143.3 _(6.9)	670.6 _(7.4)
INT-Double	$\delta_1 : -7.3_{(6.2)}$ $\delta_2 : 9.1_{(1.6)}$	170.4 _(1.1)	51.7 _(2.0)	0.50 _(0.01)	98.1_(6.6)	17.8 _(0.3)	1005.2_(3.5)	434.9_(8.3)
NERCOME	-	171.2 _(1.1)	52.1 _(2.0)	0.50 _(0.01)	99.0 _(6.7)	17.9 _(0.3)	1007.6 _(3.5)	438.9 _(8.5)
NONLIN	-	170.2_(1.1)	51.7_(2.0)	0.50 _(0.01)	99.2 _(6.5)	17.6_(0.3)	1006.2 _(3.5)	436.3 _(8.3)
Grand Avg	-	172.1 _(1.1)	52.5 _(2.1)	0.50 _(0.01)	102.1 _(6.8)	17.8 _(0.3)	1009.2 _(3.5)	445.6 _(8.8)
NOVELIST	-	174.9 _(1.2)	68.6 _(2.9)	0.51 _(0.01)	101.6 _(6.5)	18.0 _(0.3)	1020.6 _(3.9)	456.7 _(8.5)

Table 1: Mean and standard deviation (in bracket) of different losses for different methods. For POET and related methods, $K = 0$ is used. For banding, we omit the instances where the estimator is not positive definite when $p = 100$ in profile (I) and $p = 200$ in profile (It) and report the 20% trimmed mean together with its interquartile range/1.349 for L_2 , L_3 and L_7 . Gray cells indicate the minimum among all methods.

is that INT-Double, being INT-POET combined with Banding, outperforms (albeit only slightly) INT-POET in all 7 losses despite the fact that Banding is not doing good (and not even positive definite in many instances) at all. It would seem that Banding, despite its bad performance, still “contains” some tiny advantages over other estimators, and INT-Double helps extract these tiny advantages out. Incidentally, it outperforms all other methods in all 7 losses, except for NONLIN in L_3 .

Within the same profile (I), when $p = n = 200$ (so now $p/n = 1$), it is much more difficult for POET to perform well relative to NERCOME or NONLIN in terms of minimizing the Frobenius loss, hence INT-POET or INT-Double both have weight on POET reduced compared to the case when $p = 100, n = 200$. The results are closer for all the methods as well since the integrated estimators do not have as much advantages over other estimators as before as a result of POET doing worse than before relative to other estimators (except for L_3). This trend continues as we increase p from 200 to 400, when the ratio p/n becomes 2, which is difficult for regularized estimators to perform well with this relatively small n . The weight on POET for INT-POET and INT-Double decrease further, as POET itself is performing even worse than Banding on average. Hence both integrated estimators, albeit still better than NERCOME itself, is much closer to NEROME in performance over all losses. The results for $p = 200$ under profile (It) is also included in Table 1, and the pattern is similar to that for profile (I) when $p = 200$. Profile (It) for $p = 100$ and $p = 400$ are similar to the respective profile (I) for $p = 100$ and $p = 400$ too and we omit the results to save space.

From Table 2, the performance of the integrated estimators under profile (II) or (IIIt) are virtually the same as NERCOME, which is to be expected since Σ_0 has no sparsity or banded structure to be exploit, so that the corresponding weights for Banding and POET are small around 0. As such, NONLIN shows slightly better performance on average compared to NERCOME under profile (II) or (IIIt), which is also documented in Lam (2016). Profile (III) and (IIIIt) feature a Σ_0 which is constructed by adding a sparse and a banded covariance matrix together, so that Σ_0 itself is partly banded at best, and further from the diagonal it is roughly approximately sparse. Hence it is expected for the integrated estimators to put more emphasis on POET with $K = 0$, and less on Banding. This is exactly the case from Table 2, where the weight for banding is on average positive but in fact fluctuates around 0 for all integrated estimators. The emphasis on POET is much larger, although similar to profile (I) or (It) in Table 1, the weight decreases in general as p increases from 200 to 400. Under profile (III) or (IIIIt), the integrated estimators, especially INT-Double, has an edge over all other estimators on average.

Table 3 shows the results under profile (IV) and (IVt). Because of the cross non-zero pattern on Σ_0 , banding can result in gross deviation from the true values in the off-diagonal entries as long as the banding number is not very large. It is no surprise that Banding is performing very badly. INT-BAND is performing worse than NERCOME on L_3 , but is still close to NERCOME on other losses. Looking at the corresponding weight, Banding still exploits and takes advantages of the special structure of Σ_0 , but definitely not much. POET itself is also not performing particularly well, but INT-POET, having taken advantages of the sparsity of Σ_0 from POET itself, has performed well in all the losses. INT-Double suffers from the bad performance of Banding itself, and is not performing as good as INT-POET in L_3 , although it still helps INT-Double to achieve the smallest Frobenius loss L_1 overall. NONLIN and Grand Avg also have good performance overall, especially when $p = 200$. Under profile (V), with the correct number of factors specified, POET performs well overall, but INT-POET still outperforms POET in L_2 , L_6 and L_7 . The weight on Banding is virtually 0, which should be the case ideally, so that INT-Double has virtually

$n = 200$		Profile (II)						
$p = 100$	$\delta(\%)$	L_1	L_2	L_3	L_4	L_5	L_6	L_7
INT-BAND	0.4 _(3.7)	23.1 _(0.2)	15.5 _(0.3)	0.72 _(0.02)	23.5 _(1.1)	5.0 _(0.2)	132.0 _(0.8)	28.2 _(0.6)
Banding	-	36.5 _(2.6)	42.9 _(25.7)	18.53 _(22.8)	54.2 _(14.9)	8.2 _(2.6)	162.0 _(2.2)	69.6 _(4.4)
INT-POET	-0.1 _(5.2)	23.1 _(0.2)	15.5 _(0.3)	0.72 _(0.02)	23.5 _(1.1)	5.0 _(0.2)	132.0 _(0.8)	28.2 _(0.6)
POET	-	28.3 _(0.2)	23.5 _(0.4)	0.78 _(0.01)	41.0 _(0.8)	5.7 _(0.2)	146.3 _(0.8)	44.7 _(0.6)
INT-Double	$\delta_1 : 0.4$ _(3.7) $\delta_2 : -0.2$ _(5.2)	23.1 _(0.3)	15.5 _(0.3)	0.72 _(0.02)	23.5 _(1.1)	5.0 _(0.2)	132.0 _(0.8)	28.2 _(0.6)
NERCOME	-	23.1 _(0.2)	15.5 _(0.3)	0.72 _(0.02)	23.5 _(1.1)	5.0 _(0.2)	132.0 _(0.8)	28.2 _(0.6)
NONLIN	-	22.7 _(0.2)	14.9 _(0.4)	0.71 _(0.03)	21.3 _(0.7)	4.9 _(0.2)	130.4 _(0.8)	27.0 _(0.5)
Grand Avg	-	23.9 _(0.2)	18.3 _(0.2)	0.74 _(0.01)	34.0 _(0.8)	4.9 _(0.1)	136.7 _(0.8)	33.6 _(0.5)
NOVELIST	-	25.9 _(0.2)	52.4 _(1.0)	0.89 _(0.00)	37.7 _(1.0)	5.3 _(0.1)	141.0 _(0.8)	38.4 _(0.5)
$p = 200$		Profile (II _t)						
INT-BAND	-1.4 _(4.2)	39.1 _(0.3)	45.9 _(0.6)	0.78 _(0.01)	89.0 _(6.5)	5.5 _(0.2)	294.5 _(2.3)	89.1 _(2.6)
Banding	-	63.6 _(16.9)	107.3 _(94.5)	58.08 _(76.45)	140.7 _(16.4)	12.1 _(6.6)	355.4 _(25.6)	211.5 _(87.0)
INT-POET	-1.3 _(3.5)	39.2 _(0.3)	45.9 _(0.6)	0.78 _(0.01)	89.1 _(6.5)	5.5 _(0.2)	294.5 _(2.3)	89.1 _(2.5)
POET	-	47.0 _(1.8)	61.3 _(0.7)	0.81 _(0.01)	105.3 _(2.5)	7.6 _(3.6)	316.5 _(1.5)	119.6 _(2.7)
INT-Double	$\delta_1 : -1.5$ _(4.5) $\delta_2 : -0.8$ _(3.7)	39.1 _(0.3)	45.9 _(0.6)	0.78 _(0.01)	89.1 _(6.5)	5.5 _(0.2)	294.5 _(2.3)	89.1 _(2.6)
NERCOME	-	39.1 _(0.3)	45.9 _(0.6)	0.78 _(0.01)	89.0 _(6.5)	5.5 _(0.2)	294.5 _(2.3)	89.1 _(2.6)
NONLIN	-	38.9 _(2.0)	46.0 _(1.9)	0.88 _(0.12)	82.1 _(3.5)	6.2 _(3.4)	290.2 _(1.1)	86.4 _(2.4)
Grand Avg	-	40.3 _(0.2)	48.8 _(0.2)	0.78 _(0.01)	106.8 _(2.3)	5.3 _(0.1)	300.8 _(1.1)	98.5 _(0.9)
NOVELIST	-	42.8 _(1.9)	90.0 _(1.2)	0.88 _(0.00)	104.8 _(2.7)	7.0 _(3.8)	305.0 _(1.4)	104.2 _(2.7)
$p = 200$		Profile (III _t)						
INT-BAND	6.9 _(6.8)	89.6 _(1.2)	23.1 _(1.1)	0.47 _(0.03)	37.9 _(3.4)	18.2 _(0.7)	426.4 _(2.8)	148.2 _(3.9)
Banding	-	119.2 _(16.8)	40.8 _(14.4)	0.84 _(0.04)	57.4 _(8.5)	21.9 _(5.6)	492.5 _(34.3)	259.9 _(66.5)
INT-POET	15.4 _(4.2)	89.2 _(1.2)	22.9 _(1.1)	0.47 _(0.03)	37.3 _(3.4)	18.6 _(0.6)	424.5 _(3.0)	146.3 _(4.0)
POET	-	106.7 _(18.5)	31.7 _(1.1)	0.43 _(0.03)	38.1 _(3.6)	22.5 _(24.6)	462.3 _(4.9)	191.0 _(18.3)
INT-Double	$\delta_1 : 3.1$ _(10.4) $\delta_2 : 14.4$ _(5.3)	89.0 _(1.2)	22.8 _(1.1)	0.47 _(0.03)	37.2 _(3.4)	18.3 _(0.7)	424.5 _(2.9)	145.8 _(4.0)
NERCOME	-	90.0 _(1.1)	23.2 _(1.1)	0.47 _(0.03)	38.0 _(3.4)	18.7 _(0.6)	426.4 _(2.8)	148.9 _(3.9)
NONLIN	-	91.3 _(20.8)	23.0 _(1.6)	0.47 _(0.05)	37.1 _(3.3)	20.5 _(25.9)	428.2 _(3.2)	148.8 _(17.6)
Grand Avg	-	90.5 _(1.1)	23.5 _(1.2)	0.48 _(0.03)	40.8 _(3.5)	18.6 _(0.6)	427.4 _(2.9)	153.1 _(4.2)
NOVELIST	-	98.8 _(19.2)	40.3 _(2.4)	0.49 _(0.03)	41.2 _(4.2)	22.4 _(24.8)	443.6 _(4.9)	170.6 _(18.5)
$p = 400$		Profile (III _t)						
INT-BAND	3.8 _(4.6)	180.1 _(1.2)	49.9 _(1.2)	0.37 _(0.02)	87.0 _(4.0)	22.5 _(0.5)	1026.5 _(3.2)	448.1 _(6.1)
Banding	-	231.5 _(42.5)	74.6 _(18.5)	0.37 _(0.02)	113.6 _(13.1)	30.5 _(9.2)	1143.0 _(66.1)	667.7 _(133.9)
INT-POET	7.1 _(2.1)	179.8 _(1.1)	49.7 _(1.2)	0.37 _(0.02)	86.5 _(4.0)	22.7 _(0.4)	1025.0 _(3.2)	446.3 _(6.1)
POET	-	237.5 _(3.3)	94.8 _(2.1)	0.34 _(0.02)	102.2 _(3.4)	32.6 _(9.7)	1176.1 _(5.4)	715.0 _(11.4)
INT-Double	$\delta_1 : 1.2$ _(6.3) $\delta_2 : 6.7$ _(2.5)	179.7 _(1.2)	49.6 _(1.2)	0.37 _(0.02)	86.5 _(4.0)	22.5 _(0.5)	1025.0 _(3.3)	445.7 _(6.1)
NERCOME	-	180.4 _(1.1)	49.9 _(1.2)	0.37 _(0.02)	87.0 _(4.0)	22.7 _(0.4)	1026.6 _(3.2)	448.8 _(6.1)
NONLIN	-	179.7 _(1.3)	49.6 _(1.2)	0.37 _(0.02)	87.0 _(3.7)	22.7 _(2.7)	1027.9 _(3.5)	447.0 _(6.7)
Grand Avg	-	181.1 _(1.1)	50.2 _(1.2)	0.37 _(0.02)	90.2 _(4.1)	22.6 _(0.4)	1028.0 _(3.4)	455.5 _(6.8)
NOVELIST	-	194.7 _(4.5)	70.3 _(2.1)	0.37 _(0.02)	93.8 _(4.4)	29.1 _(10.5)	1064.5 _(9.6)	508.1 _(19.2)

Table 2: Mean and standard deviation (in bracket) of different losses for different methods. For POET and related methods, $K = 0$ is used. For banding, we omit each instance where the estimator is not positive definite in all profiles and report the 20% trimmed mean together with its interquartile range/1.349 for all losses. For NONLIN under profile (II_t) with $p = 200$, we do the same for L_2 and L_3 , when some instances of NONLIN are near singular. Gray cells indicate the minimum among all methods.

$n = 200$		Profile (IV)						
$p = 100$	$\delta(\%)$	L_1	L_2	L_3	L_4	L_5	L_6	L_7
INT-BAND	-25.4 _(14.1)	8.5 _(0.7)	3.9 _(0.2)	0.56 _(0.23)	3.4 _(0.2)	5.2 _(1.0)	46.0 _(1.3)	4.0 _(0.3)
Banding	-	11.7 _(1.0)	71.0 _(59.2)	48.23 _(45.37)	35.2 _(3.2)	5.0 _(0.9)	82.3 _(0.8)	18.7 _(1.7)
INT-POET	38.4 _(7.4)	8.3 _(0.7)	3.4 _(0.2)	0.35 _(0.03)	3.1 _(0.1)	5.2 _(1.0)	45.5 _(0.9)	3.6 _(0.3)
POET	-	11.2 _(0.8)	15.1 _(1.3)	2.12 _(0.24)	12.6 _(0.8)	8.0 _(1.1)	69.2 _(0.9)	9.0 _(0.4)
INT-Double	$\delta_1 : -24.8$ _(13.9) $\delta_2 : 38.3$ _(7.2)	8.3 _(0.7)	3.5 _(0.3)	0.57 _(0.29)	3.2 _(0.2)	5.2 _(1.1)	45.9 _(0.9)	3.6 _(0.3)
NERCOME	-	8.5 _(0.7)	3.9 _(0.2)	0.42 _(0.03)	3.3 _(0.1)	5.2 _(1.0)	45.6 _(1.3)	4.0 _(0.3)
NONLIN	-	8.3 _(0.6)	3.8 _(0.2)	0.42 _(0.02)	3.4 _(0.2)	4.9 _(0.9)	45.3 _(1.1)	3.9 _(0.3)
Grand Avg	-	8.6 _(0.7)	3.8 _(0.2)	0.41 _(0.02)	3.3 _(0.1)	5.4 _(1.1)	47.7 _(1.1)	4.0 _(0.3)
NOVELIST	-	10.5 _(0.7)	225.1 _(14.4)	6.56 _(0.62)	23.4 _(7.2)	4.9 _(1.1)	77.3 _(4.0)	13.5 _(3.1)
$p = 200$		Profile (IVt)						
INT-BAND	-17.5 _(30.2)	15.3 _(0.6)	12.5 _(0.8)	1.65 _(1.40)	10.5 _(1.1)	7.7 _(0.8)	100.9 _(4.3)	12.6 _(0.7)
Banding	-	22.8 _(1.0)	$> 10^3$	$> 10^3$	151.4 _(42.0)	8.3 _(1.6)	187.4 _(12.9)	71.8 _(11.7)
INT-POET	62.4 _(5.7)	14.1 _(0.7)	10.3 _(0.6)	0.48 _(0.08)	9.9 _(0.7)	7.4 _(0.9)	106.3 _(1.8)	10.5 _(0.7)
POET	-	17.3 _(0.8)	45.9 _(3.4)	3.91 _(0.55)	33.4 _(1.9)	9.5 _(0.8)	146.1 _(1.5)	23.0 _(1.0)
INT-Double	$\delta_1 : -16.6$ _(25.6) $\delta_2 : 61.0$ _(6.1)	14.0 _(0.7)	10.8 _(0.8)	1.20 _(0.95)	10.3 _(0.9)	7.4 _(0.9)	107.0 _(2.0)	10.6 _(0.7)
NERCOME	-	15.3 _(0.6)	11.9 _(0.4)	0.50 _(0.04)	9.8 _(0.6)	7.8 _(0.8)	100.0 _(4.5)	12.4 _(0.6)
NONLIN	-	15.1 _(0.7)	11.6 _(0.4)	0.50 _(0.04)	10.4 _(0.9)	7.1 _(1.0)	99.1 _(3.0)	12.4 _(0.8)
Grand Avg	-	15.5 _(0.7)	11.8 _(0.4)	0.48 _(0.04)	9.8 _(0.7)	8.1 _(0.8)	106.9 _(3.2)	12.5 _(0.7)
NOVELIST	-	21.9 _(0.7)	389.6 _(20.5)	6.59 _(0.58)	229.0 _(28.9)	7.5 _(1.2)	191.7 _(5.2)	74.4 _(6.7)
$p = 200$		Profile (V)						
INT-BAND	0.1 _(5.9)	138.2 _(8.0)	25.6 _(1.3)	0.66 _(0.04)	45.6 _(3.9)	79.2 _(12.0)	432.0 _(3.0)	172.2 _(5.1)
Banding	-	229.4 _(7.3)	$> 10^3$	$> 10^2$	153.1 _(24.0)	76.8 _(10.5)	631.0 _(9.9)	787.8 _(21.7)
INT-POET	26.4 _(5.3)	137.3 _(8.1)	24.9 _(1.2)	0.66 _(0.04)	44.2 _(3.7)	79.2 _(12.0)	428.5 _(3.2)	167.6 _(5.1)
POET	-	139.2 _(5.5)	26.8 _(1.1)	0.64 _(0.04)	42.2 _(3.3)	70.5 _(8.2)	437.0 _(3.7)	175.6 _(4.6)
INT-Double	$\delta_1 : 0.0$ _(5.7) $\delta_2 : 26.4$ _(5.3)	137.3 _(8.1)	24.9 _(1.2)	0.66 _(0.04)	44.2 _(3.7)	79.2 _(12.0)	428.5 _(3.2)	167.6 _(5.1)
NERCOME	-	138.2 _(8.0)	25.6 _(1.3)	0.66 _(0.04)	45.6 _(3.9)	79.3 _(12.0)	432.0 _(3.0)	172.2 _(5.1)
NONLIN	-	136.0 _(6.7)	25.3 _(1.3)	0.66 _(0.04)	45.0 _(3.7)	73.0 _(10.4)	432.2 _(3.0)	170.3 _(4.7)
Grand Avg	-	139.5 _(8.4)	26.0 _(1.3)	0.67 _(0.04)	49.6 _(4.3)	81.8 _(12.3)	433.1 _(3.0)	177.5 _(5.3)
NOVELIST	-	229.3 _(7.2)	255.8 _(37.7)	0.71 _(0.04)	231.0 _(3.1)	78.9 _(12.1)	629.4 _(9.0)	801.6 _(24.4)
$p = 400$		Profile (Vt)						
INT-BAND	0.0 _(1.5)	305.1 _(13.0)	55.3 _(1.9)	0.49 _(0.01)	102.5 _(6.5)	175.7 _(20.9)	1038.9 _(3.5)	533.6 _(11.3)
Banding	-	616.6 _(15.5)	-	-	-	187.6 _(44.8)	1706.8 _(80.7)	-
INT-POET	7.6 _(3.0)	304.8 _(13.0)	55.1 _(1.9)	0.49 _(0.01)	102.0 _(6.5)	175.7 _(20.9)	1037.5 _(3.5)	531.2 _(11.4)
POET	-	319.7 _(9.0)	71.7 _(1.8)	0.47 _(0.01)	103.7 _(5.6)	148.2 _(12.4)	1110.2 _(6.0)	642.9 _(14.0)
INT-Double	$\delta_1 : -0.1$ _(1.5) $\delta_2 : 7.5$ _(3.0)	304.8 _(13.0)	55.1 _(1.9)	0.49 _(0.01)	102.0 _(6.5)	175.7 _(20.9)	1037.5 _(3.5)	531.2 _(11.4)
NERCOME	-	305.0 _(13.0)	55.3 _(1.9)	0.49 _(0.01)	102.5 _(6.5)	175.7 _(20.9)	1038.8 _(3.5)	533.5 _(11.3)
NONLIN	-	300.1 _(10.1)	55.0 _(1.9)	0.49 _(0.01)	102.3 _(6.3)	159.3 _(17.8)	1040.4 _(4.1)	531.2 _(11.3)
Grand Avg	-	308.2 _(14.6)	55.7 _(2.0)	0.49 _(0.01)	106.9 _(6.8)	182.1 _(21.5)	1040.3 _(3.6)	542.3 _(12.7)
NOVELIST	-	623.5 _(13.3)	-	0.52 _(0.02)	-	179.9 _(20.5)	1738.9 _(15.5)	-

Table 3: Mean and standard deviation (in bracket) of different losses for different methods. For POET and related methods, $K = 0$ is used for profile (IV) and (IVt), and $K = 2$ is used for profile (V) and (Vt). For banding, we omit each instance where the estimator is not positive definite in all profiles and report the 20% trimmed mean together with its interquartile range/1.349 for all losses. Same for INT-Band and INT-Double under profile (IVt) with L_2 and L_3 losses. Under profile (Vt) with $p = 400$, almost all instances of Banding and NOVELIST are non-positive definite, and the corresponding L_2, L_3, L_4 and L_7 are not reported. Gray cells indicate the minimum among all methods.

the same performance as INT-POET. The same goes for profile (Vt) with $p = 400$, where the weight on Banding is virtually 0 for INT-BAND and INT-Double. The weight on POET has decreased comparing to profile (V) at $p = 200$, so that NONLIN and Grand Avg have closer performance to the integrated estimators. Again, this reflects the increased difficulties for POET to achieve a good performance as $p/n = 2$ with n relatively small at 200.

5.2 Forecasting the number of phone calls

We are interested in forecasting the number of phone calls for a call center. The data is used in Huang et al. (2006) and Bickel and Levina (2008b), and is re-analyzed in Lam (2016). Phone calls to a call center are recorded from 7am to midnight everyday in 2002, except for weekends, holiday and when equipments are malfunctioning, leaving $n = 239$ days of calls in total. A 17-hour recording period is divided into 10-minute intervals each day, resulting in 102 intervals. Let N_{ij} be the number of calls at the j th interval on the i th day, $i = 1, \dots, 239$, $j = 1, \dots, 102$. The transformation $y_{ij} = (N_{ij} + 1/4)^{1/2}$ is applied to bring the data closer to normal (see Huang et al. (2006) for more details).

We consider predicting the number of phone calls in the latter half of the day for the last 29-day period of the data using the data from the first half of the day. Lam (2016) has illustrated empirically that the last 29-day period of the data is particularly difficult to forecast, and we bring in our proposed estimators to compare to other methods used in the simulations to see if our proposed estimators can improve forecasting. In details, let $\mathbf{y}_i = ((\mathbf{y}_i^{(1)})^\top, (\mathbf{y}_i^{(2)})^\top)^\top$, where $\mathbf{y}_i^{(1)} = (y_{i,1}, \dots, y_{i,51})^\top$ and $\mathbf{y}_i^{(2)} = (y_{i,52}, \dots, y_{i,102})^\top$. Let $\boldsymbol{\mu}_j = E(\mathbf{y}_i^{(j)})$ for $j = 1, 2$, and $\boldsymbol{\Sigma}_{jk} = \text{cov}(\mathbf{y}_i^{(j)}, \mathbf{y}_i^{(k)})$, $1 \leq j, k \leq 2$. We use the best linear predictor

$$\hat{\mathbf{y}}_i^{(2)} = \boldsymbol{\mu}_2 + \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1} (\mathbf{y}_i^{(1)} - \boldsymbol{\mu}_1)$$

for predicting the number of calls in the second half of the day, and so need to estimate $\boldsymbol{\Sigma} = \text{cov}(\mathbf{y}_i)$ using past data. We consider 30, 60, 90, 120, 150, 180 and 210 days of training data prior to the 29-day period for forecasting, and for each training data set, we compare all the methods used in the simulations in Section 5.1, plus sample covariance, pure adaptive soft thresholding (SOFT, i.e., POET with $K = 0$), integrated estimator with soft thresholding (INT-SOFT), and integrated estimator with banding and soft-thresholding (INT-bandsoft). For POET, we consider $K = 1$ to explore if some parts of the data exhibits factor model structure. For banding, we consider the banding of the modified Cholesky factor instead of direct banding of the covariance matrix (see Bickel and Levina (2008b) for more details), since the components in $\mathbf{y}_i$ are time-ordered, so the dependence on distant past can be weak, meaning the modified Cholesky factor can be naturally banded.

For $k = 30, 60, \dots, 210$, we compare for different methods the absolute forecast errors

$$E_k = \frac{1}{51} \sum_{j=52}^{102} E_{k,j}, \quad \text{where } E_{k,j} = \frac{1}{29} \sum_{r=211}^{239} |\hat{y}_{rj,k}^{(2)} - y_{rj}^{(2)}|, \quad (5.1)$$

where $\hat{y}_{rj,k}^{(2)}$ is the forecast on day r and time interval j , using k prior days for constructing an estimator for $\boldsymbol{\Sigma}$. Except for SOFT, POET and NONLIN, we repeatedly estimate $\boldsymbol{\Sigma}$ for 80 times and average the forecast results to arrive at $\hat{y}_{rj,k}^{(2)}$ to reduce the variability from data splitting, and the choice of banding number for banding or the choice of thresholding parameter for NOVELIST by cross-validation.

	Training data size (days prior to the forecast period)						
	30	60	90	120	150	180	210
Sample	-	2.94 _(0.83)	1.71 _(0.50)	1.60 _(0.47)	1.59 _(0.52)	1.60 _(0.51)	1.53 _(0.49)
NERCOME	1.45 _(0.50)	1.51 _(0.47)	1.59 _(0.51)	1.58 _(0.52)	1.59 _(0.53)	1.60 _(0.55)	1.53 _(0.51)
NONLIN	1.48 _(0.52)	1.50 _(0.48)	1.60 _(0.51)	1.60 _(0.52)	1.61 _(0.54)	1.60 _(0.55)	1.53 _(0.51)
Grand Avg	1.46 _(0.50)	1.50 _(0.47)	1.59 _(0.51)	1.59 _(0.52)	1.60 _(0.53)	1.60 _(0.55)	1.54 _(0.51)
NOVELIST	1.61 _(0.50)	1.72 _(0.51)	1.64 _(0.50)	1.57 _(0.48)	1.57 _(0.52)	1.59 _(0.51)	1.52 _(0.48)
Banding	1.88 _(0.54)	1.74 _(0.51)	1.50 _(0.47)	1.47_(0.47)	1.51 _(0.51)	1.54 _(0.52)	1.44 _(0.46)
INT-BAND	1.43 _(0.49)	1.50 _(0.47)	1.58 _(0.50)	1.58 _(0.52)	1.59 _(0.53)	1.60 _(0.55)	1.53 _(0.51)
POET	1.75 _(0.55)	1.61 _(0.49)	1.71 _(0.54)	1.68 _(0.54)	1.69 _(0.56)	1.68 _(0.56)	1.54 _(0.51)
INT-POET	1.45 _(0.50)	1.50 _(0.48)	1.55 _(0.49)	1.55 _(0.51)	1.58 _(0.53)	1.59 _(0.55)	1.51 _(0.50)
INT-Double	1.42_(0.49)	1.51 _(0.48)	1.55 _(0.50)	1.56 _(0.51)	1.58 _(0.53)	1.59 _(0.55)	1.51 _(0.50)
SOFT	8.88 _(4.30)	4.81 _(1.99)	2.08 _(0.80)	1.57 _(0.46)	1.39_(0.43)	1.42_(0.45)	1.37_(0.42)
INT-SOFT	1.47 _(0.50)	1.47 _(0.38)	1.44 _(0.42)	1.48 _(0.46)	1.52 _(0.50)	1.55 _(0.53)	1.50 _(0.49)
INT-bandsoft	1.46 _(0.49)	1.45_(0.39)	1.44_(0.42)	1.48 _(0.46)	1.52 _(0.50)	1.55 _(0.53)	1.50 _(0.49)

Table 4: Mean and standard deviation (in bracket) of the $E_{k,j}$'s defined in (5.1) for different methods. Sample covariance does not have a stable Σ_{11}^{-1} for $k = 30$, and the corresponding results are omitted. Gray cells indicate the minimum for a particular training data size k .

From Table 4, the best forecasts come from soft-thresholding when we use more data ($k \geq 150$) in estimating Σ , followed closely by INT-Double and INT-bandsoft when using $k \leq 90$. Banding is good in general for larger k , which makes sense since a larger k means some components in $\mathbf{y}_i$ are further apart in time, so that the modified Cholesky factor is more banded. INT-SOFT and INT-bandsoft perform well in general for each particular k , and are in general better than the nonlinear shrinkage methods like NERCOME, NONLIN and Grand Avg. It indicates that, despite the bad performance for banding and soft-thresholding when $k \leq 60$, there are still advantages from the banding and sparse assumptions, and the corresponding integrated estimators help squeeze them out. Certainly, when k gets larger, it becomes apparent that Σ becomes sparser and with more banded structure, and banding and soft-thresholding themselves fare better.

The integrated estimators INT-SOFT and INT-bandsoft are also better than the NOVELIST in general, even when NERCOME and sample covariance are performing at the same level when $k \geq 150$. It means that the integrated estimators are better in extracting the advantages from the sparsity of Σ on average in this study.

References

- Abadir, K. M., Distaso, W., and Žikeš, F. (2014). Design-free estimation of variance matrices. *Journal of Econometrics*, 181(2):165 – 180.
- Bai, J. and Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70(1):191–221.
- Bai, Z. and Silverstein, J. (2010). *Spectral Analysis of Large Dimensional Random Matrices*. Springer Series in Statistics, New York, 2 edition.
- Bai, Z. D. and Yin, Y. Q. (1993). Limit of the smallest eigenvalue of a large dimensional sample covariance matrix. *The Annals of Probability*, 21(3):1275–1294.
- Bickel, P. J. and Levina, E. (2008a). Covariance regularization by thresholding. *The Annals of Statistics*, 36(6):2577–2604.

- Bickel, P. J. and Levina, E. (2008b). Regularized estimation of large covariance matrices. *Ann. Statist.*, 36(1):199–227.
- Cai, T. T. and Yuan, M. (2012). Adaptive covariance matrix estimation through block thresholding. *Ann. Statist.*, 40(4):2014–2042.
- Cai, T. T. and Zhou, H. H. (2012). Optimal rates of convergence for sparse covariance matrix estimation. *The Annals of Statistics*, 40(5):2389–2420.
- Fan, J., Fan, Y., and Lv, J. (2008). High dimensional covariance matrix estimation using a factor model. *Journal of Econometrics*, 147(1):186–197.
- Fan, J., Liao, Y., and Mincheva, M. (2011). High-dimensional covariance matrix estimation in approximate factor models. *The Annals of Statistics*, 39(6):3320–3356.
- Fan, J., Liao, Y., and Mincheva, M. (2013). Large covariance estimation by thresholding principal orthogonal complements. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 75(4):603–680.
- Friedman, J., Hastie, T., and Tibshirani, R. (2008). Sparse inverse covariance estimation with the graphical lasso. *Biostatistics*, 9(3):432–441.
- Huang, J. Z., Liu, N., Pourahmadi, M., and Liu, L. (2006). Covariance matrix selection and estimation via penalised normal likelihood. *Biometrika*, 93(1):85–98.
- Huang, N. and Fryzlewicz, P. (2015). Novelist estimator of large correlation and covariance matrices and their inverses. Technical report, Department of Statistics, London School of Economics and Political Science.
- Lam, C. (2016). Nonparametric eigenvalue-regularized precision or covariance matrix estimator. *Ann. Statist.*, 44(3):928–953.
- Lam, C. and Fan, J. (2009). Sparsistency and rates of convergence in large covariance matrix estimation. *The Annals of Statistics*, 37(6B):4254–4278.
- Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365 – 411.
- Ledoit, O. and Wolf, M. (2012). Nonlinear shrinkage estimation of large-dimensional covariance matrices. *The Annals of Statistics*, 40(2):1024–1060.
- Ledoit, O. and Wolf, M. (2013). Optimal estimation of a large-dimensional covariance matrix under Stein’s loss. ECON - Working Papers 122, Department of Economics - University of Zurich.
- Meinshausen, N. and Bühlmann, P. (2006). High-dimensional graphs and variable selection with the lasso. *The Annals of Statistics*, 34(3):1436–1462.
- Rothman, A. J., Levina, E., and Zhu, J. (2009). Generalized thresholding of large covariance matrices. *J. Amer. Statist. Assoc.*, 104(485):177–186.

- Schäfer, J. and Strimmer, K. (2005). A shrinkage approach to large-scale covariance matrix estimation and implications for functional genomics. *Statistical Applications in Genetics and Molecular Biology, The Berkeley Electronic Press*, 4(1).
- Won, J.-H., Lim, J., Kim, S.-J., and Rajaratnam, B. (2013). Condition-number-regularized covariance estimation. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 75(3):427–450.